

Data Analysis & Probability

ANSWER KEY



Mean, median, mode and range

- Find the mean of 6, 9, 12, 15, 18.
$$\text{Mean} = \frac{6 + 9 + 12 + 15 + 18}{5} = \frac{60}{5} = 12.$$
- Ages of 7 students: 13, 15, 13, 16, 14, 13, 17. Find the mean, median, and mode.
Ordered: 13, 13, 13, 14, 15, 16, 17. Mean = $\frac{101}{7} \approx 14.43$. Median (4th value) = 14. Mode = 13 (appears three times).
- Find the range and mode of the data set 8, 3, 8, 15, 8, 22, 3.
Range = $22 - 3 = 19$. Mode = 8 (appears three times).
- A data set has mean 18 and 6 values. Find the sum of the values.
Sum = $18 \times 6 = 108$.
- The data set 10, 12, 11, 13, 50 contains an outlier. Find the mean before and after removing 50.
Before: $\frac{10 + 12 + 11 + 13 + 50}{5} = \frac{96}{5} = 19.2$. After removing 50: $\frac{46}{4} = 11.5$.
- A data set of 5 values has mean 20. Four of the values are 15, 18, 22, 25. Find the fifth value.
Total sum = $20 \times 5 = 100$. Known sum = $15 + 18 + 22 + 25 = 80$. Fifth value = $100 - 80 = 20$.

Reading tables

- A survey of 50 people's favorite fruit: Apple 15, Banana 12, Orange 10, Mango 13. What fraction of people chose banana?
$$\frac{12}{50} = \frac{6}{25}.$$
- Using the same survey (Apple 15, Banana 12, Orange 10, Mango 13, out of 50), what percent chose either apple or mango?
 $15 + 13 = 28$ people. $\frac{28}{50} \times 100\% = 56\%.$
- A table shows quiz scores: score 5 (3 students), score 7 (5), score 8 (8), score 9 (4), score 10 (2). Find the mode score and the total number of students.
Mode = 8 (8 students, the highest frequency). Total students = $3 + 5 + 8 + 4 + 2 = 22$.
- Using the same quiz data, find the mean score to two decimal places.
Sum = $5(3) + 7(5) + 8(8) + 9(4) + 10(2) = 15 + 35 + 64 + 36 + 20 = 170$. Mean = $\frac{170}{22} \approx 7.73$.

Reading bar charts

- 11 The bar chart shows books sold by genre in one week. Find the total books sold and the modal (most common) genre.
Total = $45 + 30 + 20 + 15 = 110$ books. The modal genre is Fiction, with 45 books.
- 12 Using the same book-sales data, what percent of total books sold were mystery novels, to the nearest percent?
 $\frac{30}{110} \times 100\% \approx 27\%$.
- 13 The bar chart shows monthly rainfall in mm. Find the mean monthly rainfall.
Mean = $\frac{60 + 45 + 80 + 55}{4} = \frac{240}{4} = 60$ mm.
- 14 Using the same rainfall data (Jan 60, Feb 45, Mar 80, Apr 55 mm), find the percent decrease from March to April.
Decrease = $80 - 55 = 25$ mm. $\frac{25}{80} \times 100\% = 31.25\%$.

Basic probability

- 15 A bag holds 4 red and 6 blue marbles. Find $P(\text{red})$.
 $P(\text{red}) = \frac{4}{10} = \frac{2}{5}$.
- 16 A fair six-sided die is rolled. Find $P(\text{a number greater than 4})$.
Favorable outcomes $\{5, 6\}$: $P = \frac{2}{6} = \frac{1}{3}$.
- 17 A card is drawn from a standard deck of 52 cards. Find $P(\text{a heart})$.
 $P = \frac{13}{52} = \frac{1}{4}$.
- 18 A spinner has 8 equal sections numbered 1 to 8. Find $P(\text{an even number})$.
Even outcomes $\{2, 4, 6, 8\}$: $P = \frac{4}{8} = \frac{1}{2}$.

Combined and complementary events

- 19 A bag holds 3 red, 5 blue and 2 green marbles. Find $P(\text{not red})$.
 $P(\text{red}) = \frac{3}{10}$, so $P(\text{not red}) = 1 - \frac{3}{10} = \frac{7}{10}$.
- 20 Two fair coins are tossed. Find $P(\text{both land heads})$.
 $P = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$.
- 21 A die is rolled twice. Find $P(\text{both rolls show a 6})$.
 $P = \frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$.
- 22 A bag holds 4 white and 6 black balls. One ball is drawn and not replaced, then a second is drawn. Find $P(\text{both white})$.
 $P = \frac{4}{10} \times \frac{3}{9} = \frac{12}{90} = \frac{2}{15}$.

Weighted averages

- 23** A class of 9 students has an average score of 80. A 10th student joins with a score of 100. Find the new average for all 10 students.
Old sum = $80 \times 9 = 720$. New sum = $720 + 100 = 820$. New average = $\frac{820}{10} = 82$.
- 24** Class A has 20 students with mean score 75; Class B has 30 students with mean score 85. Find the overall mean of all 50 students.
Total sum = $20(75) + 30(85) = 1,500 + 2,550 = 4,050$. Overall mean = $\frac{4,050}{50} = 81$.
- 25** A student's final grade is 40% homework average (score 90) and 60% exam average (score 70). Find the weighted final grade.
 $0.4 \times 90 + 0.6 \times 70 = 36 + 42 = 78$.
- 26** A survey combines two groups: 15 people with average age 24, and 25 people with average age 32. Find the overall average age of all 40 people.
Total sum = $15(24) + 25(32) = 360 + 800 = 1,160$. Overall mean = $\frac{1,160}{40} = 29$.

Comparing data sets

- 27** Data set A: 10, 12, 14, 16, 18. Data set B: 8, 14, 14, 14, 20. Both have the same mean. Which has the greater range?
Both have mean 14. Range of A = $18 - 10 = 8$. Range of B = $20 - 8 = 12$. Data set B has the greater range (more spread out).
- 28** Team X scored 22, 25, 28, 25 points in 4 games. Team Y scored 20, 20, 35, 25. Both teams have the same mean score. Which team's scores are more consistent (smaller range)?
Both have mean 25. Range of X = $28 - 22 = 6$. Range of Y = $35 - 20 = 15$. Team X is more consistent.
- 29** Class 1 has 18 students with mean score 68; Class 2 has 22 students with mean score 74. Find the combined mean of all 40 students, to one decimal place.
Total sum = $18(68) + 22(74) = 1,224 + 1,628 = 2,852$. Combined mean = $\frac{2,852}{40} = 71.3$.

Word problems

- 30** A teacher records the number of siblings for 10 students: 1, 2, 0, 1, 3, 2, 1, 0, 2, 1. Find the mean, median, and mode.
Sum = 13, mean = $\frac{13}{10} = 1.3$. Ordered: 0, 0, 1, 1, 1, 1, 2, 2, 2, 3; median = $\frac{1 + 1}{2} = 1$. Mode = 1 (appears four times).

- 31 A company's weekly sales (in thousands of riyals) over 5 weeks: 40, 55, 35, 60, 50. Find the mean weekly sales, and state how many weeks had sales above the mean.
 $\text{Mean} = \frac{240}{5} = 48$ thousand riyals. Weeks above the mean: 55, 60, 50 — that is 3 weeks.
- 32 A jar contains 5 red, 3 blue and 2 yellow candies. A candy is picked and eaten, then another is picked. Find $P(\text{first red, then blue})$.
 $P = \frac{5}{10} \times \frac{3}{9} = \frac{15}{90} = \frac{1}{6}$.
- 33 In a survey of 200 students, 120 prefer math and the rest prefer science. Find the probability, as a percent, that a randomly chosen student prefers science.
 $\text{Science} = 200 - 120 = 80$ students. $P = \frac{80}{200} = 0.4 = 40\%$.

MCQ — Mean, median, mode and range

- 34 Find the mean of 8, 11, 14, 17, 20.

- a. 12
- b. 17
- c. 70
- d. 14 ✓

Sum = 70; mean = $70 \div 5 = 14$.

- 35 Ages of 6 students: 12, 14, 12, 15, 13, 12. Find the mean, median, and mode.

- a. Mean = 13, median = 12, mode = 12.5
- b. Mean = 13, median = 12.5, mode = 12 ✓
- c. Mean = 12.5, median = 13, mode = 12
- d. Mean = 13, median = 12.5, mode = 13

Sum = 78, mean = 13. Sorted: 12, 12, 12, 13, 14, 15; median = $\frac{12 + 13}{2} = 12.5$; mode = 12 (appears 3 times).

36 Find the range and mode of the data set 5, 9, 5, 18, 5, 27, 9.

a. Range = 13, mode = 5

b. Range = 27, mode = 5

c. Range = 22, mode = 5 ✓

d. Range = 22, mode = 9

Range = $27 - 5 = 22$; mode = 5 (appears 3 times).

37 A data set has mean 24 and 8 values. Find the sum of the values.

a. 32

b. 3

c. 192 ✓

d. 96

Sum = mean $\times$ count = $24 \times 8 = 192$.

38 The data set 14, 16, 15, 17, 60 contains an outlier. Find the mean before and after removing 60.

a. Before = 122, after = 15.5

b. Before = 24.4, after = 15.5 ✓

c. Before = 15.5, after = 24.4

d. Before = 24.4, after = 62

Before: sum = 122, mean = 24.4. After removing 60: sum = 62 over 4 values, mean = 15.5.

39 A data set of 6 values has mean 25. Five of the values are 20, 22, 28, 30, 32. Find the sixth value.

a. 18 ✓

b. 28

c. 8

d. 150

Total needed = $6 \times 25 = 150$; sum of five = 132; sixth = $150 - 132 = 18$.

40 The bar chart shows quiz scores. Find the total number of students, and the modal score.

a. 20 students; mode = 8 ✓

b. 20 students; mode = 6

c. 20 students; mode = 9

d. 18 students; mode = 8

Total = $2 + 4 + 6 + 5 + 3 = 20$; the score 8 has the highest frequency (6 students).

MCQ — Reading tables

41 A survey of 60 people's favorite drink: Tea 18, Coffee 22, Juice 12, Water 8. What fraction of people chose coffee?

a. $\frac{18}{60}$

b. $\frac{22}{60}$

c. $\frac{11}{30}$ ✓

d. $\frac{11}{60}$

$\frac{22}{60}$ simplifies (divide by 2) to $\frac{11}{30}$.

42 Using the same survey (Tea 18, Coffee 22, Juice 12, Water 8, out of 60), what percent chose either tea or water, to the nearest percent?

a. 43% ✓

b. 30%

c. 50%

d. 26%

$(18 + 8) \div 60 = 26 \div 60 \approx 43\%$.

- 43 A table shows test scores: score 60 (4 students), 70 (7), 80 (9), 90 (6), 100 (4). Find the total number of students.

- a. 34
- b. 30 ✓
- c. 26
- d. 25

$$4 + 7 + 9 + 6 + 4 = 30.$$

- 44 Using the same test data, find the mean score to two decimal places.

- a. 79.67 ✓
- b. 80.00
- c. 76.33
- d. 79.00

$$\text{Sum} = 240 + 490 + 720 + 540 + 400 = 2,390; \text{mean} = 2,390 \div 30 \approx 79.67.$$

- 45 A table shows the number of pets owned by 25 families: 0 pets (6 families), 1 pet (10), 2 pets (7), 3 pets (2). Find the median number of pets.

- a. 0
- b. 1.2
- c. 2
- d. 1 ✓

With 25 values sorted, the median is the 13th value, which falls in the '1 pet' group.

- 46 Using the same pet data, find the mean number of pets per family, to two decimal places.

- a. 1.00
- b. 1.20 ✓
- c. 1.50
- d. 30

$$\text{Sum} = 0(6) + 1(10) + 2(7) + 3(2) = 30; \text{mean} = 30 \div 25 = 1.20.$$

MCQ — Reading bar charts

47 The bar chart shows books sold by genre in one week. Find the total books sold, and the modal genre.

a. 120; Romance

b. 100; Fiction

c. 120; Mystery

d. 120; Fiction ✓

Total = $45 + 30 + 25 + 20 = 120$; Fiction had the most sales (45).

48 Using the same book-sales data, what percent of total books sold were mystery novels, to the nearest percent?

a. 25% ✓

b. 30%

c. 20%

d. 38%

$30 \div 120 = 25\%$.

49 The bar chart shows monthly rainfall in mm. Find the mean monthly rainfall.

a. 225

b. 65

c. 55

d. 56.25 ✓

$$\text{Mean} = \frac{50 + 70 + 40 + 65}{4} = \frac{225}{4} = 56.25 \text{ mm.}$$

- 50 Using the same rainfall data (Jan 50, Feb 70, Mar 40, Apr 65 mm), find the percent decrease from Feb to Mar, to the nearest percent.

- a. 75%
- b. 43% ✓
- c. 30%
- d. 40%

Decrease = $70 - 40 = 30$; $30 \div 70 \approx 43\%$.

- 51 The bar chart shows quarterly sales for four stores, in thousands of riyals. Find the range, and which store had the highest sales.

- a. 55; Store B
- b. 55; Store D ✓
- c. 140; Store D
- d. 35; Store D

Range = $140 - 85 = 55$; Store D had the highest sales (140).

- 52 The bar chart shows hours studied each weekday. Find the total hours, and the mean hours per day.

- a. 12; 2
- b. 12; 2.4 ✓
- c. 12; 3
- d. 10; 2

Total = $2 + 3 + 1 + 4 + 2 = 12$; mean = $12 \div 5 = 2.4$.

- 53 The bar chart shows the number of defective items found for four products in a quality check. Find the total defects, and the range.

a. 40; 10 ✓

b. 40; 7

c. 35; 10

d. 40; 15

Total = $12 + 8 + 15 + 5 = 40$; range = $15 - 5 = 10$.

MCQ — Basic probability

- 54 A bag holds 5 red and 7 blue marbles. Find $P(\text{red})$.

a. $\frac{7}{12}$

b. $\frac{5}{12}$ ✓

c. $\frac{5}{7}$

d. $\frac{1}{12}$

5 red out of 12 total.

- 55 A fair six-sided die is rolled. Find $P(\text{a number less than 3})$.

a. $\frac{1}{6}$

b. $\frac{1}{2}$

c. $\frac{2}{3}$

d. $\frac{1}{3}$ ✓

Favorable outcomes $\{1, 2\}$: $\frac{2}{6} = \frac{1}{3}$.

56 A card is drawn from a standard deck of 52 cards. Find $P(\text{a king})$.

a. $\frac{1}{4}$

b. $\frac{1}{52}$

c. $\frac{1}{13}$ ✓

d. $\frac{4}{13}$

4 kings out of 52: $\frac{4}{52} = \frac{1}{13}$.

57 A spinner has 10 equal sections numbered 1 to 10. Find $P(\text{a multiple of 3})$.

a. $\frac{1}{10}$

b. $\frac{1}{3}$

c. $\frac{4}{10}$

d. $\frac{3}{10}$ ✓

Multiples of 3 from 1–10: $\{3, 6, 9\}$, so $\frac{3}{10}$.

58 A bag holds 6 yellow and 9 green balls. Find $P(\text{yellow})$.

a. $\frac{1}{5}$

b. $\frac{6}{9}$

c. $\frac{3}{5}$

d. $\frac{2}{5}$ ✓

6 out of 15 simplifies to $\frac{2}{5}$.

59 A fair six-sided die is rolled. Find $P(\text{an odd number})$.

a. $\frac{2}{3}$

b. $\frac{1}{6}$

c. $\frac{1}{2}$ ✓

d. $\frac{1}{3}$

Odd outcomes $\{1, 3, 5\}$: $\frac{3}{6} = \frac{1}{2}$.

MCQ — Combined and complementary events

60 A bag holds 4 red, 6 blue and 5 green marbles. Find $P(\text{not blue})$.

a. $\frac{3}{5}$ ✓

b. $\frac{9}{6}$

c. $\frac{2}{5}$

d. $\frac{6}{15}$

Not blue = $4 + 5 = 9$ out of 15: $\frac{9}{15} = \frac{3}{5}$.

61 Two fair coins are tossed. Find $P(\text{at least one tail})$.

a. $\frac{3}{4}$ ✓

b. $\frac{1}{3}$

c. $\frac{1}{4}$

d. $\frac{1}{2}$

$P(\text{no tails}) = \frac{1}{4}$, so $P(\text{at least one}) = 1 - \frac{1}{4} = \frac{3}{4}$.

62 A die is rolled twice. Find $P(\text{both rolls show an even number})$.

a. $\frac{1}{6}$

b. $\frac{1}{3}$

c. $\frac{1}{4}$ ✓

d. $\frac{1}{2}$

Each roll: $P(\text{even}) = \frac{1}{2}$; both: $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$.

- 63 A bag holds 5 white and 7 black balls. One ball is drawn and not replaced, then a second is drawn. Find $P(\text{both black})$.

a. $\frac{7}{22}$ ✓

b. $\frac{6}{11}$

c. $\frac{49}{144}$

d. $\frac{7}{12}$

$$\frac{7}{12} \times \frac{6}{11} = \frac{42}{132} = \frac{7}{22}.$$

- 64 A card is drawn, replaced, then a second card is drawn from a standard deck. Find $P(\text{both hearts})$.

a. $\frac{1}{8}$

b. $\frac{1}{2}$

c. $\frac{1}{4}$

d. $\frac{1}{16}$ ✓

$$\left(\frac{13}{52}\right)^2 = \left(\frac{1}{4}\right)^2 = \frac{1}{16}.$$

- 65 A bag holds 3 red and 4 blue marbles. Two are drawn without replacement. Find $P(\text{both red})$.

a. $\frac{3}{7}$

b. $\frac{9}{49}$

c. $\frac{2}{7}$

d. $\frac{1}{7}$ ✓

$$\frac{3}{7} \times \frac{2}{6} = \frac{6}{42} = \frac{1}{7}.$$

MCQ — Weighted averages

- 66 A class of 12 students has an average score of 78. A 13th student joins with a score of 91. Find the new average.
- a. 84.50
 - b. 78.00
 - c. 91.00
 - d. 79.00 ✓

New total = $12(78) + 91 = 1,027$; new average = $1,027 \div 13 = 79.00$.

- 67 Class A has 25 students with mean score 70; Class B has 15 students with mean score 90. Find the combined mean.
- a. 80
 - b. 75
 - c. 160
 - d. 77.5 ✓

$$\frac{25(70) + 15(90)}{40} = \frac{3,100}{40} = 77.5.$$

- 68 A student's final grade is 30% homework average (score 85) and 70% exam average (score 75). Find the final grade.
- a. 82
 - b. 80
 - c. 78 ✓
 - d. 75.5

$$0.3(85) + 0.7(75) = 25.5 + 52.5 = 78.$$

- 69 A survey combines two groups: 20 people with average age 30, and 30 people with average age 40. Find the combined average age.

a. 70

b. 35

c. 36 ✓

d. 34

$$\frac{20(30) + 30(40)}{50} = \frac{1,800}{50} = 36.$$

- 70 A class of 8 students has an average score of 65. A 9th student joins, changing the new average to 68. Find the 9th student's score.

a. 71

b. 65

c. 88

d. 92 ✓

$$\text{New total} = 9(68) = 612; \text{old total} = 8(65) = 520; \text{9th score} = 612 - 520 = 92.$$

- 71 Two classes combine: Class A (10 students, mean 60) and Class B (n students, mean 90), giving a combined mean of 78 for all students. Find n .

a. 12

b. 15 ✓

c. 10

d. 20

$$\frac{600 + 90n}{10 + n} = 78 \Rightarrow 600 + 90n = 780 + 78n \Rightarrow 12n = 180 \Rightarrow n = 15.$$

MCQ — Comparing data sets

- 72 Data set A: 12, 14, 16, 18, 20. Data set B: 8, 15, 17, 19, 21. Both have the same mean. Which has the greater range?
- a. Cannot be determined
 - b. They are equal
 - c. Data set A
 - d. Data set B ✓

Both have mean 16. Range A = 8; range B = 13. B has the greater range.

- 73 Team X scored 18, 22, 25, 23 points in 4 games. Team Y scored 15, 15, 32, 26. Both teams have the same mean. Which team is more consistent (smaller range)?
- a. Team X ✓
 - b. Team Y
 - c. Cannot be determined
 - d. They are equally consistent

Both have mean 22. Range X = 7; range Y = 17. Team X is more consistent.

- 74 Class 1 has 16 students with mean score 70; Class 2 has 24 students with mean score 80. Find the combined mean of both classes.
- a. 150
 - b. 75
 - c. 76 ✓
 - d. 77

$$\frac{16(70) + 24(80)}{40} = \frac{3,040}{40} = 76.$$

- 75 Data set A: 5, 10, 15, 20, 25. Data set B: 13, 14, 15, 16, 17. Both have mean 15. Which has the smaller range, and what is it?
- a. Data set B; 20
 - b. Data set B; 4 ✓
 - c. Data set A; 4
 - d. Data set A; 20

Range A = 20; range B = 4. Data set B is smaller.

- 76 Two data sets both have median 20. Set P: 10, 15, 20, 25, 30. Set Q: 18, 19, 20, 21, 22. Which set is more spread out?
- a. Set P ✓
 - b. Cannot be determined
 - c. Set Q
 - d. They are equally spread out

Range P = 20; range Q = 4. Set P is more spread out.

MCQ — Word problems

- 77 A teacher records the number of siblings for 12 students: 2, 1, 0, 2, 3, 1, 2, 0, 1, 2, 3, 1. Find the mean, to two decimal places.
- a. 1.33
 - b. 1.60
 - c. 1.50 ✓
 - d. 2.00

Sum = 18; mean = $18 \div 12 = 1.50$.

- 78 A company's weekly sales (in thousands of riyals) over 6 weeks: 45, 60, 38, 72, 55, 50. Find the mean weekly sales.
- a. 55.00
 - b. 320
 - c. 50.00
 - d. 53.33 ✓

Sum = 320; mean = $320 \div 6 \approx 53.33$.

- 79 A jar contains 6 red, 4 blue and 2 yellow candies. A candy is picked and eaten, then another is picked. Find $P(\text{both red})$.

- a. $\frac{6}{12}$
- b. $\frac{5}{22}$ ✓
- c. $\frac{1}{4}$
- d. $\frac{3}{11}$

$$\frac{6}{12} \times \frac{5}{11} = \frac{30}{132} = \frac{5}{22}.$$

- 80 In a survey of 300 students, 180 prefer math and the rest prefer science. Find the probability, as a percent, that a randomly chosen student prefers science.

- a. 120%
- b. 40% ✓
- c. 18%
- d. 60%

Science = 120 out of 300: 40%.

- 81 The bar chart shows a company's quarterly sales, in thousands of riyals. Find the total annual sales, and the mean quarterly sales.

- a. 1,560; 450
- b. 1,560; 390 ✓
- c. 1,560; 400
- d. 1,200; 390

Total = $320 + 410 + 380 + 450 = 1,560$; mean = $1,560 \div 4 = 390$.

- 82 A survey of 150 shoppers found 90 prefer online shopping and the rest prefer in-store. Find the probability, as a fraction in simplest form, that a randomly chosen shopper prefers in-store.

- a. $\frac{2}{5}$ ✓
- b. $\frac{90}{150}$
- c. $\frac{3}{5}$
- d. $\frac{3}{50}$

In-store = 60 out of 150, which simplifies to $\frac{2}{5}$.

- 83 A basketball player makes 18 out of 25 free throws attempted. Find the probability, as a decimal, that the next free throw is missed (based on this rate).

- a. 0.28 ✓
- b. 0.18
- c. 0.25
- d. 0.72

Missed rate = $\frac{25 - 18}{25} = \frac{7}{25} = 0.28$.